ITA0612 – MACHINE LEARNING

CO2-AT3

SCENARIO BASED QUESTONS

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QUESTION:1

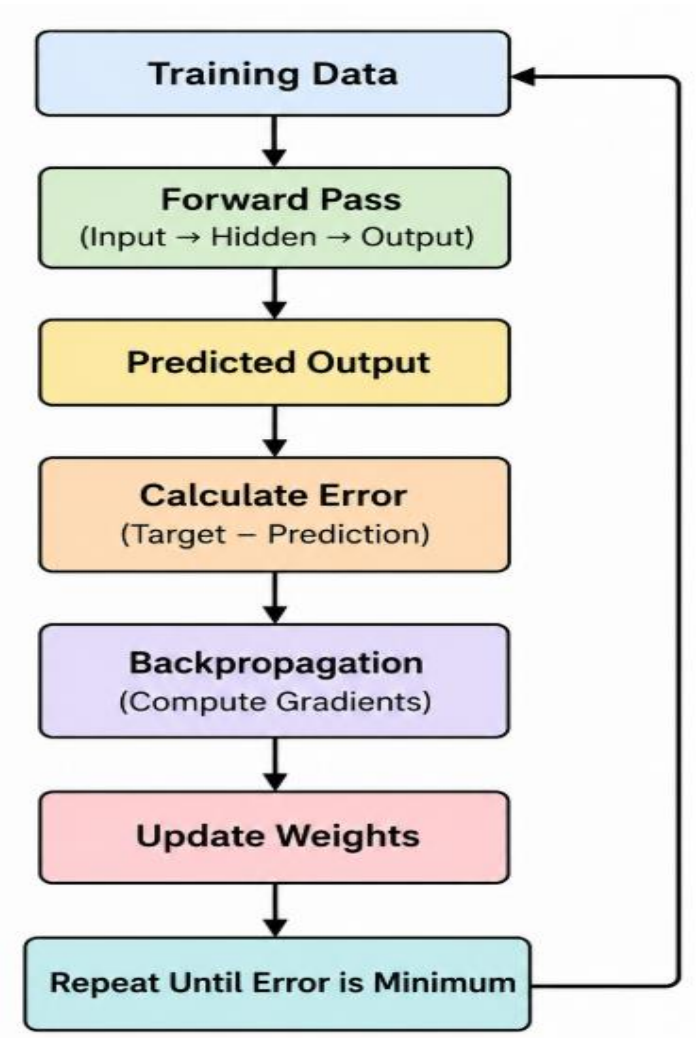
In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.

ANSWER:

Introduction

A Multilayer Perceptron (MLP) is a feedforward artificial neural network consisting of an input layer, one or more hidden layers, and an output layer. For binary classification, the output neuron predicts either 0 or 1. The backpropagation algorithm is the learning process that adjusts the network's weights to minimize prediction errors.

Architecture of an MLP



Step 1: Forward Pass

The input data passes through the network.

Example

Suppose:

- Input:
    - $x_1 = 1$
    - $x_2 = 0$
- Target Output = 1

Initial weights:

- $x_1 \rightarrow h_1 = 0.4$
- $x_2 \rightarrow h_1 = 0.6$
- $h_1 \rightarrow$ Output = 0.5

Hidden Neuron Calculation

$$Net\_h = (1 \times 0.4) + (0 \times 0.6)$$

$$Net\_h = 0.4$$

Using the **Sigmoid activation function:**

Hidden Output

$$= 1 / (1 + e^{-0.4})$$

$$\approx 0.598$$

Output Layer

$$Net\_o = 0.598 \times 0.5$$

$$= 0.299$$

Applying Sigmoid:

$$\text{Predicted Output} \approx 0.574$$

The network predicts 0.574, while the expected output is 1.

Step 2: Error Calculation

The prediction error is calculated using the loss function.

Mean Squared Error (MSE):

Error

$= \frac{1}{2}(\text{Target} - \text{Prediction})^2$

$= \frac{1}{2}(1 - 0.574)^2$

≈ 0.091

Since the error is not zero, the weights need to be updated.

Step 3: Backward Propagation

The error is propagated backward through the network.

The gradient (partial derivative) tells:

- Which weight caused the error
- How much each weight should change

Weight Update Formula

$$\text{New Weight} = \text{Old Weight} - \eta \times \text{Gradient}$$

where

- η = Learning Rate

Role of Learning Rate

The **learning rate (η)** controls how much the weights change during each update.

Small Learning Rate

- Slow learning
- Stable convergence
- More epochs required
- Less chance of overshooting

Large Learning Rate

- Faster learning
- May overshoot the minimum error
- Can become unstable
- May fail to converge

Example:

$\eta = 0.01$

Very slow learning $\eta = 0.1$

Balanced learning (commonly used) $\eta = 1.0$

Too large (May diverge)

Role of Activation Functions

Activation functions introduce non-linearity, allowing the network to learn complex patterns

Activation Function	**Formula**	**Advantages**	**Disadvantages**
Sigmoid	$1/(1+e^{-x})$	Good for binary classification; outputs values between 0 and 1	Vanishing gradient problem; slower convergence
Tanh	$(e^{x}-e^{-x})/(e^{x}+e^{-x})$	Zero-centered outputs; better than Sigmoid for hidden layers	Vanishing gradient problem
ReLU	max(0,x)	Fast training; computationally efficient; avoids saturation for positive values	Dead neuron problem for negative inputs
Leaky ReLU	max(0.01x,x)	Reduces dead neuron problem; improves gradient flow	Slightly higher computational cost than ReLU

Effect on Convergence and Accuracy

Learning Rate

- Appropriate learning rate → Faster convergence and better accuracy.
- Too small → Slow convergence.
- Too large → Oscillation or divergence.

Activation Function

- ReLU speeds up training.
- Sigmoid provides probability outputs for binary classification.
- Proper activation functions improve model accuracy and convergence.

Advantages of Backpropagation

- Efficient weight optimization.
- Learns complex nonlinear relationships.
- Reduces prediction error iteratively.
- Works with multiple hidden layers.
- Widely used in deep learning applications.

Conclusion:

Backpropagation is the core learning algorithm of a multilayer perceptron. It performs a forward pass to compute predictions, calculates the prediction error, and then propagates this error backward to update the network weights using gradient descent. The learning rate determines the size of weight updates, while activation functions enable the network to model nonlinear relationships. Together, these components help the neural network converge to an accurate solution for binary classification tasks.

QUESTION:2

In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. Choose your own dataset. Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

ANSWER:

Introduction

A Genetic Algorithm (GA) is an optimization technique inspired by the process of natural selection. It starts with a population of candidate solutions and repeatedly applies selection, crossover, and mutation to evolve better solutions. The objective is to maximize a fitness function while finding the most suitable candidate.

Scenario

Suppose a company wants to identify the best AI model configuration based on:

- **Speed (%)** – Higher is better
- **Accuracy (%)** – Higher is better

Fitness Function

Fitness = (0.4 × Speed) + (0.6 × Accuracy)

Accuracy is given a higher weight because prediction quality is more important.

Initial Dataset (Generation 0)

Candidate	Chromosome	Speed	Accuracy	Fitness	Suitable
A	1010	70	80	76	Yes
B	1100	85	65	73	No
C	0110	75	90	84	Yes
D	1001	60	70	66	No

Initial Population

Population

A → 1010 → Fitness = 76

B → 1100 → Fitness = 73

C → 0110 → Fitness = 84

D → 1001 → Fitness = 66

The highest fitness is **Candidate C (84)**.

Step 1: Selection

Selection chooses individuals with **higher fitness values** to become parents.

Selected Parents

Parent	Fitness
C	84
A	76

These two candidates are selected because they have the highest fitness values.

Step 2: Crossover

Perform **Single-Point Crossover** after the second bit.

Parent Chromosomes

Parent 1 **0110**

Parent 2 **1010**

Cut after the second bit.

01 | 10

10 | 10

Exchange the last halves.

Offspring

Child 1 **0110**

Child 2 **1010**

To better demonstrate variation, assume crossover occurs after the first bit.

0 | 110

1 | 010

New offspring become

Child 1 = **0010**

Child 2 = **1110**

Assume their evaluated performance is:

Child	Speed	Accuracy	Fitness
0010	78	82	80.4
1110	88	91	89.8

The second child has a higher fitness than both parents.

Step 3: Mutation

Mutation randomly changes one gene to maintain diversity.

Before Mutation **1110**

Flip the last bit.

After Mutation **1111**

Assume the mutated chromosome improves performance.

Candidate	Speed	Accuracy	Fitness
1111	90	94	92.4

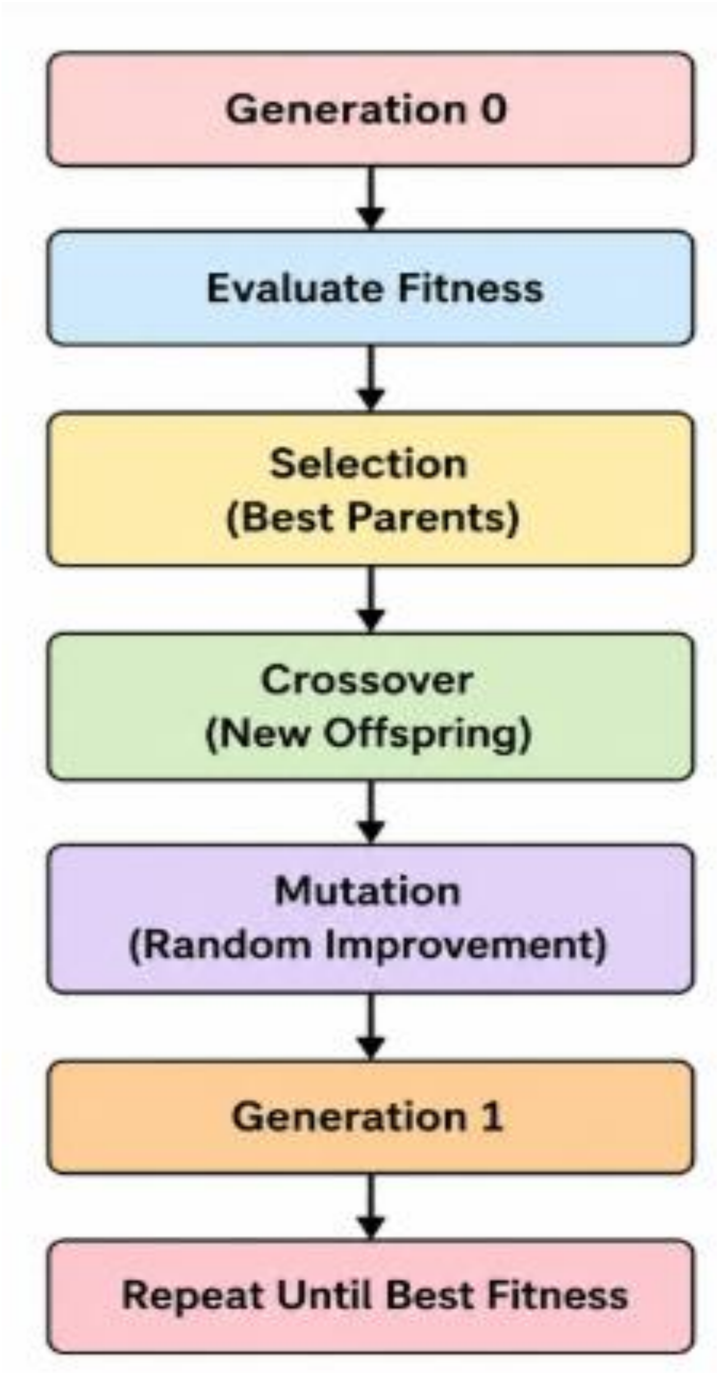
This becomes the best solution.

Generation-wise Evolution

Generation	**Best Chromosome**	**Fitness**
Generation 0	**0110**	**84.0**
Generation 1	**1110**	**89.8**
Generation 2	**1111**	**92.4**

Fitness increases in every generation, showing successful evolution.

Evolution Process



Explanation of Genetic Operations

1. Selection

- Evaluates the fitness of every candidate.
- Candidates with higher fitness are more likely to be selected as parents.
- Poor-performing candidates have a lower chance of reproduction.

- Ensures that good characteristics are passed to the next generation.

Example: Candidates C (84) and A (76) are selected because they have the highest fitness values.

2. Crossover

- Combines genes from two selected parents.
- Produces new offspring with characteristics inherited from both parents.
- Increases the possibility of generating better solutions.

Example:

Parent 1 : 0110

Parent 2 : 1010

↓

Child 1 : 0010

Child 2 : 1110

The offspring **1110** achieves a higher fitness than either parent.

3. Mutation

- Randomly changes one or more genes.
- Prevents the algorithm from getting trapped in local optima.
- Introduces new genetic diversity into the population.

Example:

Before Mutation 1110

↓

After Mutation 1111

The mutation improves the fitness from **89.8** to **92.4**.

Improvement Across Generations

Generation	Average Fitness	Best Fitness
Generation 0	74.75	84.0
Generation 1	82.50	89.8
Generation 2	88.20	92.4

The population becomes progressively stronger as evolution continues.

Final Optimized Solution

Best Chromosome	Speed	Accuracy	Fitness	Suitable
1111	90%	94%	92.4	Yes

Advantages of Genetic Algorithms

- Finds near-optimal solutions efficiently.
- Handles complex optimization problems.
- Does not require gradient information.
- Explores a large search space through mutation.

Conclusion

In this scenario, the Genetic Algorithm began with four candidate solutions and improved them through selection, crossover, and mutation. Selection chose the fittest individuals, crossover combined their best characteristics, and mutation introduced diversity to discover even better solutions. Over successive generations, the fitness improved from 84.0 to 92.4, resulting in the optimized chromosome 1111, which achieved 90% Speed, 94% Accuracy, and was identified as the best suitable solution

QUESTION:3

In the context of Neural Networks for classification, consider the given dataset used to predict whether a patient has a disease based on Age, Blood Pressure, Glucose Level, and Lifestyle Risk. Explain how a neural network model can be trained using backpropagation. Compute the values of TP, TN, FP, and FN. Evaluate the model using suitable metrics. Also explain how preprocessing and feature selection improve the model's performance.

ANSWER:

GIVEN DATASET:

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

✓ **Neural Network Training using Backpropagation**

A **Neural Network** learns the relationship between the patient attributes (**Age, Blood Pressure, Glucose Level, and Lifestyle Risk**) and the disease status (**Yes/No**).

Step 1: Input Layer

The input features are:

- Age
- Blood Pressure
- Glucose Level
- Lifestyle Risk

These are fed into the neural network.

Age

Blood Pressure

Glucose Level

Lifestyle Risk

|

▼

Input Layer

Step 2: Hidden Layer

The hidden layer extracts patterns from the inputs using weighted connections and an activation function (ReLU or Sigmoid).

Input Layer

|

▼

Hidden Layer

Step 3: Output Layer

The output neuron predicts

- Yes (Disease Present)
- No (Disease Absent)

using the **Sigmoid activation function**.

Step 4: Error Calculation

The predicted output is compared with the actual output.

If prediction ≠ actual value, an error (loss) is calculated.

Example:

Actual = Yes

Predicted = No

↓

Error = High

Step 5: Backpropagation

The error is propagated backward.

Weights are updated using

New Weight = Old Weight − Learning Rate × Gradient

This process repeats for multiple epochs until the prediction error becomes minimum.

✓ **Confusion Matrix**

Treat **"Yes"** as the Positive class.

Values

- True Positive (TP) = 2
- True Negative (TN) = 1
- False Positive (FP) = 1
- False Negative (FN) = 1

Confusion Matrix

	Predicted Yes	Predicted No
Actual Yes	TP = 2	FN = 1
Actual No	FP = 1	TN = 1

✓ **Performance Evaluation Metrics**

(a) Accuracy

$$\text{Accuracy} = \frac{\boldsymbol{TP + TN}}{\boldsymbol{TP + TN + FP + FN}}$$

$$= \frac{\mathbf{2 + 1}}{\mathbf{2 + 1 + 1 + 1}} = \frac{\mathbf{3}}{\mathbf{5}} = \mathbf{0.60}$$

Accuracy = 60%

(b) Precision

$$\text{Precision} = \frac{\boldsymbol{TP}}{\boldsymbol{TP + FP}}$$

$$= \frac{\mathbf{2}}{\mathbf{2 + 1}} = \frac{\mathbf{2}}{\mathbf{3}} = \mathbf{0.667}$$

Precision = 66.7%

(c) Recall (Sensitivity)

$$\text{Recall} = \frac{\boldsymbol{TP}}{\boldsymbol{TP + FN}}$$

$$= \frac{\mathbf{2}}{\mathbf{2 + 1}} = \frac{\mathbf{2}}{\mathbf{3}} = \mathbf{0.667}$$

Recall = 66.7%

(d) F1-Score

$$\boldsymbol{F1} = \frac{\mathbf{2} \times \boldsymbol{Precision} \times \boldsymbol{Recall}}{\boldsymbol{Precision + Recall}}$$

$$= \frac{\mathbf{2 \times 0.667 \times 0.667}}{\mathbf{0.667 + 0.667}} = \mathbf{0.667}$$

F1-Score = 66.7%

✓ **Summary of Evaluation Metrics**

Metric	Formula	Value
True Positive (TP)	Correct Yes predictions	2
True Negative (TN)	Correct No predictions	1
False Positive (FP)	Incorrect Yes prediction	1
False Negative (FN)	Incorrect No prediction	1
Accuracy	(TP + TN) / Total	60%
Precision	TP / (TP + FP)	66.7%
Recall	TP / (TP + FN)	66.7%
F1-Score	2PR / (P + R)	66.7%

✓ **How Preprocessing Improves Accuracy**

Before training the neural network, the dataset should be preprocessed.

(i) Handle Missing Values

- Replace missing values with mean, median, or mode.
- Prevents incorrect learning.

(ii) Encode Categorical Variables

Convert:

- High → 2
- Medium → 1
- Low → 0
- Yes → 1
- No → 0

This allows the neural network to process categorical data.

(iii) Feature Scaling

Normalize numerical features such as Age.

Example:

Age

30–60

↓

0–1

Scaling speeds up convergence and improves training stability.

(iv) Remove Noise

Incorrect or duplicate records are removed, improving prediction quality.

- ✓ **How Feature Selection Improves Reliability**

Feature selection chooses the most relevant attributes.

For this dataset, useful features include:

- ✓ Age
- ✓ Blood Pressure
- ✓ Glucose Level
- ✓ Lifestyle Risk

Benefits:

- Removes irrelevant features.
- Reduces overfitting.
- Speeds up training.
- Improves prediction accuracy.
- Makes the model more reliable and easier to interpret.

Conclusion

The neural network is trained using backpropagation, where patient features are passed through the network, prediction errors are calculated, and weights are updated iteratively to minimize the loss. From the given dataset, the confusion matrix values are TP = 2, TN = 1, FP = 1, and FN = 1. The model achieves 60% Accuracy, 66.7% Precision, 66.7% Recall, and 66.7% F1-Score. Proper data preprocessing (handling missing values, encoding categorical features, and feature scaling) together with feature selection improves the model's accuracy, convergence speed, and overall reliability.